Aryan Naik

aryannaik1205@gmail.com | github.com/RobotsAndAryan | London, UK

MEng Robotics and Artificial Intelligence, University College London (UCL) | Expected 2029

EDUCATION

University College London (UCL) | MEng Robotics and Artificial Intelligence | 2nd Year | Expected 2029

Relevant Coursework: Microcontroller programming, electronics hardware integration, circuit design, embedded systems fundamentals, signal processing, linear algebra, programming in C

PROJECTS

Job Drop AI Agent | Python, LangGraph, AWS Bedrock, SQLite, RAG

- Reduced manual job-tracking overhead to zero across a five-stage interview pipeline (applied, phone screen, technical, final, outcome) by building a multi-agent LangGraph state machine with SQLite persistence and Human-in-the-Loop graph checkpointing via Telegram notifications
- Improved application triage accuracy by implementing sentence-transformer RAG with cosine similarity scoring to rank job postings scraped from Reed.co.uk and Gradcracker across multiple role categories
- Delivered ATS-optimised PDF output for every application cycle by automating CV and cover letter compilation via pdfkit, reducing per-application document preparation time to under one minute
- Increased agent decision quality over successive runs by injecting historical interview outcome data as context into AWS Bedrock Claude Sonnet 4.6 LLM inference calls, enabling self-improving retrieval behaviour

Robotics Hardware Challenge | C, Microcontroller Programming, Autodesk Fusion 360, Breadboard Circuit Design

- Delivered a functional electromechanical prototype within a fixed-duration hardware challenge by integrating breadboard circuit wiring, microcontroller logic programmed in C, and a mechanically constrained CAD assembly modelled in Autodesk Fusion 360
- Validated correct sensor-actuator behaviour across all required test cases by debugging embedded microcontroller logic at the register level and iterating on circuit wiring until all functional acceptance criteria were met

SKILLS

Languages: Python, C

Embedded and Hardware: Microcontroller programming, breadboard circuit integration, register-level debugging, electronics hardware integration, CAD (Autodesk Fusion 360)

Software and Frameworks: LangGraph, AWS Bedrock, SQLite, pdfkit, BeautifulSoup4, sentence-transformers

Concepts: Signal processing, state machine design, RAG, cosine similarity scoring, Human-in-the-Loop workflows, graph checkpointing, embedded systems fundamentals

ADDITIONAL

- Actively developing embedded and low-level programming skills through UCL coursework in microcontroller architecture and circuit design
- Open to firmware, DSP, and mixed-signal adjacent internship projects aligned with Cirrus Logic's embedded software focus

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Hiring Team Cirrus Logic

Dear Hiring Team,

I am writing to apply for the 2027 Software Internship at Cirrus Logic. Cirrus Logic's reputation for precision mixed-signal and audio semiconductor engineering, and its consistent investment in software-driven innovation across its product lines, make it a compelling environment for an engineer who wants to work at the intersection of low-level hardware and intelligent software systems. I am a second-year MEng Robotics and Artificial Intelligence student at University College London, graduating in 2029, and I am eager to contribute to a team where software quality directly shapes the performance of real-world silicon.

My most relevant project is a Python-based multi-agent AI system I designed and built independently. The system orchestrates a LangGraph state machine backed by AWS Bedrock Claude for LLM inference, a retrieval-augmented generation pipeline using sentence-transformers and cosine similarity scoring, and a SQLite database tracking a full application pipeline across multiple stages. I integrated BeautifulSoup4 web scraping across live job boards, a Telegram human-in-the-loop approval workflow, and automated ATS-optimised PDF generation via pdfkit. The result is a self-improving, end-to-end automated agent that reduces manual job-search overhead significantly while maintaining human oversight at critical decision points. This project sharpened my ability to architect reliable, modular software systems under real constraints.

During this placement I would bring strong Python engineering skills, experience designing stateful systems, and a methodical approach to debugging and integration. I am particularly keen to develop my understanding of embedded software development and how software interfaces with hardware at the driver and firmware level, areas central to Cirrus Logic's work that would meaningfully advance my engineering foundation.

Yours sincerely, Aryan Naik